Separable Rank of Radial Kernels in Two and Higher Dimensions: Machine-Checked Proofs in Lean 4 and Isabelle/HOL

GitHub Copilot

Source archive (CC0 1.0): https://github.com/OlliNiemitalo/kernel-rank-lean

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Abstract

A d-variate kernel $K : X_0 \times \cdots \times X_{d-1} \to \mathbb{K}$ has *separable rank* at most N if it is a sum of N coordinatewise products $\sum_{n=0}^{N-1} \prod_{j=0}^{d-1} f_{n,j}(x_j)$. On a d-dimensional Cartesian grid of support width L, a rank-R kernel evaluates in dRL one-dimensional taps per grid point rather than L^d. While a positive Gaussian $\exp(-\alpha\|x\|^2)$ has separable rank 1, approximating flat-top or oscillatory isotropic kernels—such as defocus disks, Airy point-spread functions, circular low-pass filters, or band-limited wave propagators—with finite coordinatewise separable rank relies on signed or oscillatory profiles built from complex-rate Gaussians and exponential polynomials.

We give a complete machine-checked characterization in Lean 4 (with Mathlib) of the d-way separable rank of continuous circularly and spherically symmetric kernels $K(x) = \varphi(\|x\|^2)$ on $\mathbb{R}^d$ for all $d \geq 2$: (1) separableRank$(K) < \infty$ over $\mathbb{R}$ if and only if φ agrees on $[0, \infty)$ with a well-formed exponential-polynomial representation $\sum_i \mathrm{Re}(P_i(s)e^{\lambda_i s})$ (nonzero polynomials P_i, distinct rates λ_i with $\mathrm{Im}\,\lambda_i \geq 0$, and $P_i \in \mathbb{R}[X]$ when $\mathrm{Im}\,\lambda_i = 0$); (2) over $\mathbb{C}$, a continuous radial profile $\psi(\|x\|^2)$ has exact d-way separable rank $\sum_{\mu \in S}((d-1)\deg P_\mu + 1)$ for nonzero polynomials P_μ at distinct rates; (3) over $\mathbb{R}$, a continuous radial profile $\varphi(\|x\|^2)$ has exact d-way separable rank $\sum_i c_d(P_i, \lambda_i)$ where each real-rate component ($\mathrm{Im}\,\lambda_i = 0$) costs $(d-1)\deg P_i + 1$ real d-way products (achieved by real Lagrange interpolation) and each nonreal conjugate-pair component ($\mathrm{Im}\,\lambda_i > 0$) costs $R_i = 2(d-1)\deg P_i + d$ real d-way products, achieved via an alternating R_i-point root-of-unity polynomial-product identity (giving d rather than 2^{d-1} real products for a complex Gaussian when $\deg P_i = 0$) and proved optimal by Alder–Strassen degree peeling over $\mathbb{R}$; and (4) in $d = 2$, scalar extension $\mathbb{R} \hookrightarrow \mathbb{C}$ preserves rank unconditionally, whereas for $d \geq 3$ it can strictly lower rank (e.g. $\mathrm{Re}(e^{i\|x\|^2})$ has complex rank 2 and real rank d, and $\|x\|^2\cos(\|x\|^2)$ in $d = 3$ has complex rank 6, real border rank at most 6, and exact real rank 7); moreover, in $d = 2$ the exact rank equals the degree of the minimal monic constant-coefficient half-line differential annihilator $p(D)\varphi_{\mathbb{C}} = 0$ ($p \in \mathbb{C}[X]$).

Necessity reduces to showing that a continuous one-parameter semigroup $M : [0, \infty) \to \mathcal{A}$ is an exponential $M(t) = \exp(tA)$. Alongside the 96-line Banach-algebra integral-smoothing proof used in the main development, we formalize four finite-dimensional alternatives—one by finite differences in Lean 4 and three by Euler's infinitesimal step in Lean 4 and Isabelle/HOL using a homogeneous common-scale normalization $\eta = \varepsilon + \|M(\varepsilon) - 1\|$ that avoids any *a priori* difference-quotient bound—and compare their lengths, library reliance, and generality. Finally, we connect the root-of-unity identity to planar harmonic ("beach-ball") differential operators and evaluate two GPU-verified numerical applications on an NVIDIA RTX 4070 Ti Super: (i) minimax 2-D circular lens blur, showing how nuclear-norm-optimal $\mathrm{GL}_N(\mathbb{R})$ rebalancing on rowSpan(K) eliminates up to 1,110× intermediate cancellation so `float16` buffers stay within 0.16/255 of `float64` (in contrast to $d \geq 3$, where Kruskal's theorem forbids continuous basis rotations), and (ii) explicit 3-D acoustic wave-equation leapfrog propagators, identifying Kowalczyk and van Walstijn's 27-point IWB

stencil as the single-real-Gaussian ($R = 1$) baseline and designing certified von Neumann-stable multi-Gaussian propagators ($R = 4$–9) that cut directional phase-velocity error up to 0.80π from 9.23%–19.71% to 0.055% (381×–529× faster wall-clock and 731× less memory than 27-point IWB at a 0.1% dispersion tolerance).

1 Introduction

Convolving a d-dimensional array with a dense $L \times \cdots \times L$ spatial kernel requires L^d multiply–accumulate operations per output sample. When the kernel $K : \mathbb{R}^d \to \mathbb{K}$ admits a rank-R coordinatewise decomposition

$$K(x_0, \ldots, x_{d-1}) = \sum_{n=0}^{R-1} \prod_{j=0}^{d-1} f_{n,j}(x_j), \tag{1}$$

the d-dimensional convolution reduces to R cascades of d one-dimensional convolutions, costing dRL operations per sample (or $O(dR)$ operations independent of L when the one-dimensional factors admit recursive causal/anti-causal realizations). The least $R \in \mathbb{N} \cup \{\infty\}$ for which (1) holds is the *separable rank* of K: matrix rank when $d = 2$, and functional canonical polyadic (CP) tensor rank when $d \geq 3$.

For circularly or spherically symmetric (*radial*) kernels in dimension $d \geq 2$,

$$K(x) = \varphi\left(\sum_{j=0}^{d-1} x_j^2\right), \qquad x = (x_0, \ldots, x_{d-1}) \in \mathbb{R}^d,$$

Maxwell's characterization of the normal distribution implies that a continuous nonvanishing radial kernel of separable rank 1 has a Gaussian profile $\varphi(s) = \alpha e^{\lambda s}$. Real-rate Gaussians ($\lambda < 0$), however, are strictly monotone and convex in $s = \|x\|^2$; nonnegative combinations of them cannot exhibit flat passbands, sharp cutoffs, or zero crossings. Gaussian mixtures among radial kernels. In image processing, optics, and numerical wave propagation, O. Niemitalo [17, 20, 21] and related filter-design work [13] approximate non-Gaussian radial profiles—such as circular defocus disks, Airy diffraction patterns $(J_1(r)/r)^2$, and ideal circular low-pass filters $\omega_c J_1(\omega_c r)/(2\pi r)$—by sums of *complex-rate* Gaussians $\mathrm{Re}(\beta e^{(-\alpha+i\omega)\|x\|^2})$, whose damped radial oscillations $\cos(\omega\|x\|^2 + \theta)$ synthesize sharp transitions and sidelobe rings while keeping finite separable rank.

This raises three mathematical questions:

1. **Classification ($d \geq 2$):** Must *every* continuous radial profile $\varphi : [0, \infty) \to \mathbb{R}$ of finite d-way separable rank be an exponential polynomial in $s = \|x\|^2$?
2. **Exact d-way rank over $\mathbb{C}$ and over $\mathbb{R}$:** Given an exponential-polynomial or multi-Gaussian profile with distinct rates, what is the exact minimum number of d-way product terms in (1)? Can algebraic recombinations beat naive real-and-imaginary expansion, and does complexification preserve separable rank beyond two dimensions?
3. **Proof architecture in interactive theorem provers:** In formalizing necessity—whose analytic core is the continuous one-parameter semigroup theorem $M(t) = \exp(tA)$ on $[0, \infty)$—can an elementary infinitesimal proof in the style of Euler be made simpler than classical integral smoothing once formalized in Lean 4 or Isabelle/HOL?

1.1 Contributions and overview

All numbered theorems, propositions, and lemmas in Sections 2–4 are machine-checked without unproved steps (`sorry` in Lean 4, oracles in Isabelle/HOL) in the open-source repository

`kernel-rank-lean`, using Lean `v4.36.0-rc2` with Mathlib `v4.36.0-rc2` and Isabelle2026; Section 5 (along with the border-rank remark in Section 3) develops analytical corollaries for harmonic differential operators and continuous-domain pass schedules:

- **Unified d-dimensional classification (Theorem 3.3, `separableRank_radial_lt_top_iff_exists`):** For every $d \geq 2$ and every $\varphi : \mathbb{R} \to \mathbb{R}$ continuous on $[0, \infty)$, the radial kernel $x \mapsto \varphi(\sum_{j=0}^{d-1} x_j^2)$ has finite d-way separable rank if and only if φ agrees on $[0, \infty)$ with a canonical real exponential-polynomial representation $\sum_i \mathrm{Re}(P_i(s)e^{\lambda_i s})$.
- **Exact d-D complex radial rank (Theorem 3.1, `separableRank_radial_complex_eq_iff`):** For $d \geq 2$ and $\psi : \mathbb{R} \to \mathbb{C}$ continuous on $[0, \infty)$, $\mathrm{separableRank}(x \mapsto \psi(\|x\|^2)) = N$ if and only if $\psi(s) = \sum_{\mu \in S} P_\mu(s)e^{\mu s}$ on $[0, \infty)$ for distinct rates $\mu \in S$ and nonzero polynomials P_μ satisfying $\sum_{\mu \in S}((d-1)\deg P_\mu + 1) = N$. The upper bound uses shifted root-of-unity Fourier extraction on truncated exponential polynomials; the lower bound uses multi-mode Alder–Strassen degree peeling.
- **Exact d-D real radial rank for all degrees (Theorem 3.5, `separableRank_radial_eq_iff`):** For every $d \geq 2$, $\mathrm{separableRank}(x \mapsto \varphi(\|x\|^2)) = N$ over $\mathbb{R}$ if and only if φ agrees on $[0, \infty)$ with a well-formed representation whose d-dimensional cost (`multiTotalCost`) equals N, counting $(d-1)\deg P_i + 1$ for real rates ($\mathrm{Im}\,\lambda_i = 0$, via real Lagrange interpolation) and $R_i = 2(d-1)\deg P_i + d$ for nonreal rates ($\mathrm{Im}\,\lambda_i > 0$, via an alternating root-of-unity polynomial-product identity `re_coeff_prod_eq_sum_re` and Alder–Strassen peeling over $\mathbb{R}$). Specializing to $\deg P_i = 0$ gives $r + dc$ (d real terms rather than 2^{d-1} per complex Gaussian, `separableRank_radial_gaussian_profile`), and specializing to $d = 2$ recovers totalCost (`separableRank_radial_two_eq_iff`) alongside the minimal differential-annihilator characterization (Theorem 3.6).
- **Comparative formalization of the semigroup step (Section 4):** We compare the 96-line Banach-algebra integral-smoothing proof with a 256-line finite-difference proof in Lean 4 and three finite-dimensional nonstandard proofs (206 lines in Lean via ultrafilter sequences, 399 lines in Lean via an ultrapower interface, and 275 lines in Isabelle/HOL), all simplified by the homogeneous common-scale normalization $\eta = \varepsilon + \|M(\varepsilon) - 1\|$.
- **Steerable operators, 2-D lens blur, and 3-D acoustic propagation (Section 5):** Specializing the root-of-unity identity to equal directional-derivative factors recovers odd and even planar "beach-ball" harmonic operators $\mathrm{Re}((\partial_x + i\partial_y)^m)$ [19, 22] and pass-count bounds for steerable derivative banks [18]. Complementing the formal proofs, reproducible numerical scripts and GPU benchmarks in `Applications` evaluate (i) 2-D circular disc blur across 18 component families, showing how $\mathrm{GL}_N(\mathbb{R})$ rebalancing on $\mathrm{rowSpan}(K)$ removes 595×–1,110× channel cancellation for `float16` rendering, and (ii) explicit 3-D acoustic wave propagators certified over the 3-D Brillouin zone, reducing isotropic dispersion up to 0.80π by two orders of magnitude over 7-point and 27-point FDTD schemes [10].

1.2 Related work

Radial characterizations, Gaussian sums, and wave propagators. Maxwell's kinetic-theory argument (1860) and Schoenberg's characterization of completely monotone radial functions [26] identify single Gaussians and positive Gaussian mixtures among radial kernels. For oscillatory or compactly supported radial kernels in scientific computing, Beylkin and Monzón [3] approximate profiles and Green's functions by short sums of complex exponentials and Gaussians; in image filtering, Lampert and Wirjadi [13] study anisotropic Gaussian separations, while Niemitalo [17, 20, 21] fits circular defocus and low-pass profiles directly with complex-variance Gaussians. In numerical acoustics and seismic wave propagation, Kowalczyk and

van Walstijn [10] and Savioja and Svensson [25] analyze 27-point compact explicit FDTD stencils to mitigate grid dispersion, while Fomel, Ying, and Song [6] approximate Fourier-domain wave-extrapolation symbols by low-rank separable expansions; Section 5.3 bridges these two approaches in physical space via certified von Neumann-stable radial multi-Gaussian stencils. None of the prior works establishes a machine-checked converse for arbitrary signed finite-rank radial kernels or exact d-way real CP ranks across higher-degree components.

Functional equations, Hankel operators, and real vs. complex tensor rank. Reducing radial kernels via $s_j = x_j^2$ turns K into the d-way additive Hankel kernel $\varphi(s_0 + \cdots + s_{d-1})$. In two variables, functional equations of the form $f(x+y) = \sum_{n=1}^{N} X_n(x)Y_n(y)$ go back to Levi-Civita [15], Stone [27], Anselone and Korevaar's theorem on finite-dimensional translation-invariant subspaces [2], and Laczkovich [12], alongside Kronecker's theorem (1881) on finite-rank discrete Hankel matrices; our continuous half-line formalization uses operator-norm continuous shift semigroups on $\mathcal{W}(\varphi)$ to avoid differentiability hypotheses beyond continuity. In $d \geq 3$ variables, symmetric and Hankel tensor decompositions go back to Sylvester's apolarity theory (1851) and Comon and Mourrain [4]; Kruskal [11] established generic essential uniqueness of minimal three-way CP decompositions, Landsberg [14] treats multilinear rank substitution (originating in Alder and Strassen [1]) and border rank, while Reznick [24] and Qi, Comon, and Lim [23] show that real symmetric/CP rank can strictly exceed complex rank. Our Theorems 3.1 and 3.5 give exact closed-form complex and real d-way ranks for the entire continuous Hankel exponential-polynomial class, pinning down the exact real-over-complex gap ($\mathrm{cost}_{d,\mathbb{R}} - \mathrm{cost}_{d,\mathbb{C}} = d-2$ for each nonreal conjugate pair, independent of polynomial degree).

Formalized analysis and nonstandard analysis. Banach-space calculus, Bochner integration, and normed-algebra exponentials in Lean's Mathlib [16] provide the infrastructure for `Semigroup.lean`, while Isabelle/HOL's multivariate analysis library [8] and Fleuriot's formalization of nonstandard analysis [5] (`HOL-Nonstandard_Analysis`) underpin `Semigroup_NSA.thy`. Section 4 provides a controlled side-by-side comparison of standard and nonstandard formalizations of the same theorem across both systems.

2 Definitions, Surjections, and the Two-Dimensional Core

2.1 Multi-way separable rank and exponential-polynomial representations

Throughout the paper, $\varphi : \mathbb{R} \to \mathbb{R}$ denotes a real-valued radial profile and $\psi : \mathbb{R} \to \mathbb{C}$ denotes a complex-valued radial profile. Let ι be a finite index type and $(X_i)_{i\in\iota}$ a family of sets, and let $\mathbb{K}$ be a commutative semiring. In `Defs.lean`:

Definition 2.1 (`IsSumOfProducts`, `separableRank`)**.** A kernel $K : (\prod_{i\in\iota} X_i) \to \mathbb{K}$ is a *sum of N products* if there exist univariate factors $f : \mathrm{Fin}\, N \to \prod_{i\in\iota}(X_i \to \mathbb{K})$ such that

$$K(x) = \sum_{n\in \mathrm{Fin}\, N} \prod_{i\in\iota} f_n(i, x_i) \qquad \text{for all } x \in \prod_{i\in\iota} X_i.$$

Its *separable rank* is $\mathrm{separableRank}(K) = \inf\{N \in \mathbb{N} : \mathrm{IsSumOfProducts}(K, N)\} \in \mathbb{N} \cup \{\infty\}$ (`ENat`, written `Nat`-with-top in Mathlib).

Separable rank is subadditive under pointwise sums (`IsSumOfProducts.finsetSum`) and monotone under coordinatewise composition (`IsSumOfProducts.comp`). Crucially, coordinatewise *surjections* preserve separable rank exactly (`separableRank_comp_surjective`): if $g_i : Y_i \to X_i$ has a right inverse for each $i \in \iota$, then

$$\mathrm{separableRank}\big(y \mapsto K(i \mapsto g_i(y_i))\big) = \mathrm{separableRank}(K). \tag{2}$$

Since $x \mapsto x^2$ maps $\mathbb{R}$ onto $\mathbb{R}_{\geq 0} = [0, \infty)$, applying (2) to $\iota = \operatorname{Fin} d$ identifies the d-way separable rank of any radial kernel with that of the d-way additive Hankel kernel (`separableRank_radial_eq_hankel` in `Translations.lean`):

$$\operatorname{separableRank}\left(x \in \mathbb{R}^d \mapsto \varphi\left(\sum_{j=0}^{d-1} x_j^2\right)\right) = \operatorname{separableRank}\left(s \in \mathbb{R}^d_{\geq 0} \mapsto \varphi\left(\sum_{j=0}^{d-1} s_j\right)\right). \tag{3}$$

The same lemma immediately transfers every theorem below to axis-aligned ellipsoidal kernels $\varphi(\sum_j x_j^2/a_j^2)$ $(a_j \neq 0)$ and ℓ_p superellipsoidal kernels $\varphi(\sum_j |x_j|^p)$ $(p > 0)$. Moreover, for $d \geq 2$, padding $(y_0, y_1) \in \mathbb{R}^2$ with $d-2$ zeros (`padTwo`) yields the slice monotonicity bound (`separableRank_radial_two_le`):

$$\operatorname{separableRank}\big(y \in \mathbb{R}^2 \mapsto \varphi(y_0^2 + y_1^2)\big) \leq \operatorname{separableRank}\left(x \in \mathbb{R}^d \mapsto \varphi\left(\sum_{j=0}^{d-1} x_j^2\right)\right). \tag{4}$$

To state real-valued classifications without redundant conjugate duplicates, a *component* $c = (P, \lambda)$ (`Component`) pairs a polynomial $P \in \mathbb{C}[X]$ with a complex rate $\lambda \in \mathbb{C}$ and evaluates on $\mathbb{R}$ as $c.\operatorname{eval}(s) = \operatorname{Re}(P(s)e^{\lambda s})$, with profile $\operatorname{profile}(\mathrm{cs}, s) = \sum_{c \in \mathrm{cs}} c.\operatorname{eval}(s)$.

Definition 2.2 (`Component.WellFormed`, `IsRepresentation`, `multiTotalCost`)**.** A component $c = (P, \lambda)$ is *well-formed* if $P \neq 0$, $\operatorname{Im}\lambda \geq 0$, and $\operatorname{Im}\lambda = 0 \implies P \in \mathbb{R}[X]$ (every coefficient of P has zero imaginary part). A list cs is a *representation* (`IsRepresentation`) if all its components are well-formed and their rates λ are pairwise distinct. In dimension $d \geq 2$, the d-way rendering cost of c (`Component.multiCost`) is

$$c.\operatorname{multiCost}(d) = \begin{cases} (d-1)\deg P + 1 & \text{if } \operatorname{Im}\lambda = 0, \\ 2(d-1)\deg P + d & \text{if } \operatorname{Im}\lambda > 0, \end{cases}$$

with list sum $\operatorname{multiTotalCost}(d, \mathrm{cs}) = \sum_{c \in \mathrm{cs}} c.\operatorname{multiCost}(d)$. At $d = 2$, $c.\operatorname{multiCost}(2) = c.\operatorname{cost}$ ($\deg P + 1$ for real rates, $2(\deg P + 1)$ for nonreal rates, summed in `totalCost`). A representation is *Gaussian* (`IsGaussianRepresentation`) if every $c \in \mathrm{cs}$ has $\deg P = 0$, where $\operatorname{multiTotalCost}(d, \mathrm{cs}) = \operatorname{gaussianTotalCost}(d, \mathrm{cs}) = r + dc$ for r real and c nonreal components.

Remark 2.3 (Bijection with conjugate-symmetric complex exponential polynomials)**.** Why do the conditions $\operatorname{Im}\lambda \geq 0$ and $(\operatorname{Im}\lambda = 0 \implies P \in \mathbb{R}[X])$ make real representations canonical? Because $\operatorname{Re}(\overline{P}(s)e^{\bar{\lambda}s}) = \operatorname{Re}(P(s)e^{\lambda s})$ (where $\overline{P}$ conjugates coefficients) and $\operatorname{Re}(iae^{\mu s}) = 0$ for $a, \mu \in \mathbb{R}$, unrestricted pairs (P, λ) over-parameterize real profiles. Under `Component.WellFormed`, each real rate ($\operatorname{Im}\lambda = 0, P \in \mathbb{R}[X]$) expands in $\mathbb{C}$ to a single term $P(s)e^{\lambda s}$, whereas each upper-half-plane rate ($\operatorname{Im}\lambda > 0$) expands via $\operatorname{Re} z = \frac{1}{2}(z + \bar{z})$ to the conjugate pair $\frac{1}{2}P(s)e^{\lambda s} + \frac{1}{2}\overline{P}(s)e^{\bar{\lambda}s}$ at distinct rates $\lambda \neq \bar{\lambda}$ (`injective_rates_with_conjugates`). Since exponential polynomials at distinct complex rates are $\mathbb{C}$-linearly independent on $(0, \infty)$ (`eq_zero_of_sum_expPoly_eq_zero'` in `ExpPoly.lean`), a continuous real profile determines its unordered set of well-formed components uniquely. Notice also that for $\operatorname{Im}\lambda > 0$, the complex rank of this conjugate pair by Theorem 3.1 is $2((d-1)\deg P + 1) = 2(d-1)\deg P + 2$, so $c.\operatorname{multiCost}(d) - 2((d-1)\deg P + 1) = d - 2$: every nonreal conjugate pair incurs an exact real-over-complex rank penalty of $d - 2$, independent of $\deg P$.

2.2 Two-coordinate row span, complexification, and basis-free necessity

When $\iota = \operatorname{Fin} 2$ and $\mathbb{K}$ is a field, `RowSpan.lean` defines $\operatorname{rowSpan}(K) \leq (X \to \mathbb{K})$ as the $\mathbb{K}$-linear span of the sections $\{x \mapsto K(x, y) : y \in X\}$ (fixing the second coordinate and varying

the first) and proves that $\mathrm{separableRank}(K) < \infty$ holds if and only if $\mathrm{rowSpan}(K)$ is finite-dimensional over $\mathbb{K}$, in which case $\mathrm{separableRank}(K) = \mathrm{finrank}_{\mathbb{K}}(\mathrm{rowSpan}\, K)$ (`separableRank_eq_finrank_rowSpan`). For the two-variable Hankel kernel $(s_0, s_1) \mapsto \varphi(s_0 + s_1)$ on $\mathbb{R}^2_{\geq 0}$, the row span is literally the *translate span*

$$\mathcal{W}(\varphi) = \mathrm{span}_{\mathbb{K}}\{u \in \mathbb{R}_{\geq 0} \mapsto \varphi(u + t) : t \in \mathbb{R}_{\geq 0}\} \leq (\mathbb{R}_{\geq 0} \to \mathbb{K}).$$

Scalar extension $\mathbb{R} \hookrightarrow \mathbb{C}$ (`Complexification.lean`) gives the bound $\mathrm{separableRank}_{\mathbb{C}}(K_{\mathbb{C}}) \leq \mathrm{separableRank}_{\mathbb{R}}(K)$ for every finite index set ι, where $K_{\mathbb{C}}(x) = (K(x) : \mathbb{C})$ (`separableRank_ofReal_le`). In *two* coordinates ($\iota = \mathrm{Fin}\, 2$), the reverse inequality also holds (`separableRank_ofReal`): $\mathbb{R}$-linearly independent rows in $\mathbb{R}^X$ remain $\mathbb{C}$-linearly independent in $\mathbb{C}^X$ (tested by taking real and imaginary parts of any $\mathbb{C}$-linear relation), so $\mathrm{finrank}_{\mathbb{C}}\, \mathrm{rowSpan}(K_{\mathbb{C}}) = \mathrm{finrank}_{\mathbb{R}}\, \mathrm{rowSpan}(K)$, including when both ranks are ∞.

Necessity and the two-dimensional lower bound share a single coordinate-free realization in `TranslationSemigroup.lean`:

1. Suppose $\psi : \mathbb{R} \to \mathbb{C}$ is continuous on $[0, \infty)$ and $\dim_{\mathbb{C}} \mathcal{W}(\psi) < \infty$. Equipping $\mathcal{W}(\psi)$ with a norm transferred across a linear equivalence $\mathcal{W}(\psi) \simeq_{\mathbb{C}} \mathbb{C}^n$ makes $\mathcal{W}(\psi)$ a finite-dimensional Banach space whose induced topology agrees with the subspace topology inherited from the product topology on $\mathbb{R}_{\geq 0} \to \mathbb{C}$ (since all Hausdorff vector topologies on $\mathbb{C}^n$ coincide).
2. Because $\mathcal{W}(\psi)$ is invariant under non-negative shifts $(T_t f)(u) = f(u+t)$ $(t \geq 0)$, continuity of ψ gives pointwise continuity of $t \mapsto T_t f$ on each generator and hence operator-norm continuity of the semigroup $t \mapsto T_t \in \mathcal{L}(\mathcal{W}(\psi))$ on $[0, \infty)$ (`continuous_shiftCLM`).
3. Applying the Banach-algebra semigroup theorem (`Semigroup.lean`, Section 4) yields a generator $A \in \mathcal{L}(\mathcal{W}(\psi))$ with $T_t = \exp(tA)$ for all $t \geq 0$, so $\psi(t) = \mathrm{eval}_0(\exp(tA)\psi_0)$ where $\psi_0 = \psi|_{\mathbb{R}_{\geq 0}} \in \mathcal{W}(\psi)$ (`profile_of_generator`).
4. Decomposing ψ_0 into generalized eigenspaces $\ker((A - \mu I)^k)$ of A (`Module.End.iSup_maxGenEigenspace_eq_top` in `Necessity.lean`) truncates the exponential series of $(A - \mu I)$ on each summand, expressing $\psi(t) = \sum_{\mu \in S} Q_\mu(t) e^{\mu t}$ for $t \geq 0$. When $\psi = \varphi_{\mathbb{C}}$ comes from a real profile φ, averaging $\varphi(t) = \frac{1}{2}(\varphi(t) + \overline{\varphi(t)})$ and pairing μ with $\bar{\mu}$ using exponential-polynomial uniqueness on $(0, \infty)$ (`ExpPoly.lean`, `RealExpPoly.lean`) regroups the sum into a well-formed real representation (`Representation.lean`).

For the two-dimensional exact rank and annihilator characterization (`Differential.lean`, `ComplexRank.lean`, `Annihilator.lean`), let D denote differentiation within $[0, \infty)$ (right differentiation at 0). Applying Cayley–Hamilton to the generator A on $\mathcal{W}(\psi)$ shows that the monic characteristic polynomial $\chi_A(X)$ of degree $\dim_{\mathbb{C}} \mathcal{W}(\psi)$ annihilates every orbit $t \mapsto \ell(\exp(tA)v)$, so $\chi_A(D)\psi = 0$ on $[0, \infty)$. Conversely, for $\psi(t) = \sum_i P_i(t) e^{\lambda_i t}$ with distinct rates λ_i and $P_i \neq 0$, Hasse-derivative independence shows that $q(D)\psi = 0$ on $[0, \infty)$ if and only if the product annihilator

$$p(X) = \prod_i (X - \lambda_i)^{\deg P_i + 1}$$

divides $q(X)$ (`differentialOperator_sum_expPoly_eq_zero_iff`). Thus $p \mid \chi_A$, giving $\sum_i (\deg P_i + 1) = \deg p \leq \deg \chi_A = \dim_{\mathbb{C}} \mathcal{W}(\psi)$, while Taylor shift expansion of $P_i(u+t)e^{\lambda_i(u+t)}$ gives the matching upper bound $\dim_{\mathbb{C}} \mathcal{W}(\psi) \leq \sum_i (\deg P_i + 1)$.

3 Exact Multi-Way Separable Ranks in $d \geq 2$ Dimensions

In dimension $d \geq 3$, there is no single row-span dimension equal to $\mathrm{separableRank}(K)$, and scalar extension $\mathbb{R} \hookrightarrow \mathbb{C}$ no longer preserves rank. Instead, `MultiComplexRank.lean` and

`MultiRealRank.lean` establish exact d-way upper bounds by root-of-unity and Lagrange identities and matching lower bounds by mapping functional decompositions onto finite-dimensional multilinear forms.

3.1 Bridging function spaces and multilinear tensor rank

Suppose V is a $\mathbb{K}$-vector space over a field $\mathbb{K}$, $v : \alpha \to V$ is an evaluation map whose image spans V ($\mathrm{span}_{\mathbb{K}}(\mathrm{range}\, v) = \top$), and $M : V^d \to \mathbb{K}$ is a d-multilinear map (`MultilinearMap` $\mathbb{K}$ (Fin $d \to V$) $\mathbb{K}$). Lemma `exists_linear_factors_of_span_eq_top` proves that if the composed kernel $s \in \alpha^d \mapsto M(v(s_0), \ldots, v(s_{d-1}))$ admits a functional decomposition (1) of rank R with arbitrary set-theoretic factors $f_{n,j} : \alpha \to \mathbb{K}$, then there exist *linear* functionals $\ell_{n,j} \in V^*$ such that

$$M(u_0, \ldots, u_{d-1}) = \sum_{n=0}^{R-1} \prod_{j=0}^{d-1} \ell_{n,j}(u_j) \qquad \text{for all } u_0, \ldots, u_{d-1} \in V. \tag{5}$$

Indeed, selecting a $\mathbb{K}$-basis $\{v(s_b)\}_{b \in B}$ of V from range(v) defines unique linear functionals $\ell_{n,j} \in V^*$ satisfying $\ell_{n,j}(v(s_b)) = f_{n,j}(s_b)$; both sides of (5) are d-multilinear maps that agree on every d-tuple of basis vectors, so they agree on V^d. To verify $\mathrm{span}_{\mathbb{K}}(\mathrm{range}\, v) = \top$ when $\dim_{\mathbb{K}} V < \infty$, Lemma `span_range_eq_top_of_dual` shows by quotient duality that it suffices to check $L \circ v = 0 \implies L = 0$ for every $L \in V^*$.

3.2 Exact d-way complex exponential-polynomial rank

Fix $d \geq 2$, a finite index set ι, injective rates $\lambda : \iota \to \mathbb{C}$, and nonzero polynomials $P_i \in \mathbb{C}[X]$ of degree $k_i = \deg P_i$ ($i \in \iota$).

Upper bound via shifted root-of-unity interpolation. For a single polynomial $P \in \mathbb{C}[X]$ of degree $k = \deg P$, put $L = (d-1)k + 1 \geq 1$ and $\zeta = \exp(2\pi i/L)$. Using the q-th Hasse derivative $D^{[q]}P$ (so $P(s_0 + t) = \sum_{q=0}^{k} (D^{[q]}P)(s_0)\, t^q$), define the reversed Hasse polynomial on coordinate 0 and the truncated exponential polynomials on coordinates $j \in \{1, \ldots, d-1\}$ (`revHassePoly`, `truncExpPoly`):

$$R(s_0; X) = \sum_{q=0}^{k} q!\, (D^{[q]}P)(s_0)\, X^{k-q}, \qquad E_k(s_j; X) = \sum_{q=0}^{k} \frac{s_j^q}{q!} X^q,$$

$$Q(X) = R(s_0; X) \prod_{j=1}^{d-1} E_k(s_j; X).$$

Binomial convolution gives $[X^q] \prod_{j=1}^{d-1} E_k(s_j; X) = (s_1 + \cdots + s_{d-1})^q / q!$ for all $0 \leq q \leq k$, so the coefficient of X^k in $Q(X)$ is precisely $P(s_0 + s_1 + \cdots + s_{d-1})$. Crucially, because $d \geq 2$, we have $k < L$ and $\deg Q \leq k + (d-1)k = k + L - 1 < k + L$. For *any* polynomial Q with $\deg Q < k + L$ and $k < L$, discrete Fourier averaging over only L roots of unity extracts $[X^k]Q(X)$ (`coeff_eq_sum_rootsOfUnity`):

$$[X^k]Q(X) = \frac{1}{L} \sum_{r=0}^{L-1} \zeta^{-rk}\, Q(\zeta^r),$$

because every monomial index $n \in \{0, \ldots, k+L-1\}$ satisfies $-L < -k \leq n-k < L$, so $L \mid (n-k)$ holds if and only if $n = k$. Multiplying the j-th factor of $Q(\zeta^r)$ by $e^{\lambda s_j}$ (and absorbing $\frac{1}{L}\zeta^{-rk}$ into coordinate 0) expresses $P(\sum_{j=0}^{d-1} s_j) e^{\lambda \sum_{j=0}^{d-1} s_j}$ as a sum of $L = (d-1)\deg P + 1$ coordinatewise products (`isSumOfProducts_expPoly`).

Lower bound via Alder–Strassen degree peeling. Let $m_0(i) = \deg P_i + 1$ and form the coefficient evaluation space $V = \prod_{i\in\iota} \mathbb{C}^{m_0(i)}$ with $v(s)_{i,q} = \frac{s^q}{q!}e^{\lambda_i s}$ $(0 \le q < m_0(i))$. By univariate exponential-polynomial uniqueness (`eq_zero_of_sum_expPoly_eq_zero'`), no nonzero functional in V^* vanishes on range(v), so $\operatorname{span}_{\mathbb{C}}(\operatorname{range} v) = \top$ (`span_expMonomial_eq_top`). Under v, the Hankel kernel corresponds to the d-multilinear truncated-convolution form (`truncForm`) with cutoffs $c_i \le m_0(i)$,

$$\mathcal{T}_c(u^{(0)},\dots,u^{(d-1)}) = \sum_{i\in\iota} \sum_{q_0+\cdots+q_{d-1}<c_i} w_i(q_0+\cdots+q_{d-1}) \prod_{j=0}^{d-1} u^{(j)}_{i,q_j},$$

where $w_i(c_i-1) \neq 0$ whenever $c_i > 0$. Lemma `truncForm_rank_ge` proves

$$\sum_{i:\, c_i>0} \big((d-1)(c_i-1)+1\big) \le R$$

for every rank-R decomposition (5) by strong induction on $S = \sum_{i\in\iota} c_i$:

- *Case $c_{i_0} = 1$:* Evaluating $\mathcal{T}_c$ at $(e_{i_0,0},\dots,e_{i_0,0})$ gives $w_{i_0}(0) \neq 0$, selecting an index n_0 with $\ell_{n_0,0}(e_{i_0,0}) \neq 0$. Let $\pi_{\neq i_0} : V \to V$ zero out block i_0. Substituting $\varphi_0(x) = \pi_{\neq i_0}(x) - \frac{\ell_{n_0,0}(\pi_{\neq i_0}(x))}{\ell_{n_0,0}(e_{i_0,0})} e_{i_0,0}$ on mode 0 and $\pi_{\neq i_0}$ on modes $j \ge 1$ annihilates $\ell_{n_0,0}\circ\varphi_0$ while reducing c_{i_0} from 1 to 0 (since $d \ge 2$ zeroes block i_0 on mode 1), leaving at most $R-1$ terms.
- *Case $c_{i_0} \ge 2$:* Set $s_{\mathrm{top}} = e_{i_0,c_{i_0}-1}$ and $s_0 = e_{i_0,0}$ $(s_{\mathrm{top}} \neq s_0)$. Because $(c_{i_0}-1)+(c_{i_0}-1) \ge c_{i_0}$, any rank-1 perturbation of the form $\psi(x) = x - \gamma(x)s_{\mathrm{top}}$ leaves $\mathcal{T}_c$ unchanged whenever another coordinate equals s_{top}. Stepping through modes $j = 0,\dots,d-2$ sequentially evaluates $\mathcal{T}_c$ with $u^{(j)} = s_{\mathrm{top}}$ and $u^{(k)} = \psi_k(s_0)$ $(k \neq j)$ to find $d-1$ distinct indices $D \subseteq \operatorname{Fin} R$ $(|D| = d-1)$ killed on modes $0,\dots,d-2$ by rank-1 perturbations ψ_j along s_{top}. On modes $0,\dots,d-2$ we then truncate block i_0 to indices $q < c_{i_0}-1$ before applying ψ_j, while on mode $d-1$ we shift block i_0's degree indices up by $+1$ $(e_{i_0,q} \mapsto e_{i_0,q+1}$ for $q < c_{i_0}-1)$: this transforms $\mathcal{T}_c$ into the reduced form $\mathcal{T}_{c'}$ $(c' = c[i_0 \mapsto c_{i_0}-1])$ with shifted weights $w'_{i_0}(m) = w_{i_0}(m+1)$—preserving the nonzero top weight $w'_{i_0}(c'_{i_0}-1) = w_{i_0}(c_{i_0}-1) \neq 0$—while dropping all $d-1$ terms in D.

Combining this lower bound at $c = m_0$ with the upper bound and Section 2 gives:

Theorem 3.1 (d-D Complex Radial Rank, `separableRank_radial_complex_eq_iff`)**.** *Let $d \ge 2$ and let $\psi : \mathbb{R} \to \mathbb{C}$ be continuous on $[0,\infty)$. For every $N \in \mathbb{N}$,*

$$\operatorname{separableRank}\left(x \in \mathbb{R}^d \mapsto \psi\left(\sum_{j=0}^{d-1} x_j^2\right)\right) = N$$

if and only if there exist a finite set $S \subset \mathbb{C}$ and polynomials $(P_\mu)_{\mu\in S}$ with $P_\mu \neq 0$ such that $\psi(s) = \sum_{\mu\in S} P_\mu(s)e^{\mu s}$ for all $s \ge 0$ and $\sum_{\mu\in S}\big((d-1)\deg P_\mu + 1\big) = N$.

3.3 Exact d-way real rank for arbitrary degrees (`MultiRealRank.lean`)

Before treating polynomial degrees, note that for constant factors $(d \ge 1)$, expanding $\operatorname{Re}(\prod_{j=0}^{d-1} z_j)$ into real and imaginary parts gives 2^{d-1} real products, whereas d real products always suffice:

Proposition 3.2 (Alternating $2d$-th root-of-unity identity, `re_prod_eq_sum_re`)**.** *For every integer $d \ge 1$ and all $z_0,\dots,z_{d-1} \in \mathbb{C}$,*

$$\operatorname{Re}\left(\prod_{j=0}^{d-1} z_j\right) = \frac{2^{d-1}}{d} \sum_{m=0}^{d-1} (-1)^m \prod_{j=0}^{d-1} \operatorname{Re}\left(z_j e^{-i\pi m/d}\right). \tag{6}$$

Proof. Write $\mathrm{Re}(z_j e^{-i\pi m/d}) = \frac{1}{2}\big(\overline{z_j} e^{i\pi m/d} + z_j e^{-i\pi m/d}\big)$ and expand over subsets $S \subseteq \{0,\dots,d-1\}$: multiplying by $(-1)^m = e^{i\pi m}$ yields the geometric progression w_S^m with $w_S = e^{2\pi i|S|/d}$ ($w_S^d = 1$), whose sum over $0 \le m < d$ equals d for $S \in \{\emptyset, \{0,\dots,d-1\}\}$ and 0 for $0 < |S| < d$. □

Absorbing $\frac{2^{d-1}}{d}(-1)^m$ into coordinate 0 shows that the real part of a rank-N complex d-way kernel has real rank at most Nd (`IsSumOfProducts.re`), establishing qualitative d-D classification:

Theorem 3.3 (d-D Finite-Rank Classification, `separableRank_radial_lt_top_iff_exists`)**.** *Let $d \ge 2$ and let $\varphi : \mathbb{R} \to \mathbb{R}$ be continuous on $[0,\infty)$. Then*

$$\mathrm{separableRank}\left(x \in \mathbb{R}^d \mapsto \varphi\left(\sum_{j=0}^{d-1} x_j^2\right)\right) < \infty$$
$$\iff \exists\, \mathrm{cs},\ \mathrm{IsRepresentation}(\mathrm{cs}) \wedge \forall s \ge 0,\ \varphi(s) = \mathrm{profile}(\mathrm{cs}, s).$$

For degree $k = \deg P \ge 1$, however, multiplying Theorem 3.1's complex rank $(d-1)k+1$ by d via `IsSumOfProducts.re` gives $d(d-1)k+d$, which is non-optimal. Instead, sharp upper bounds differ for real and nonreal rates (`isSumOfProducts_radial_multiComponent`):

- *Real rates ($\mathrm{Im}\,\lambda = 0$, $P \in \mathbb{R}[X]$ of degree k):* On coordinates $1,\dots,d-1$, the real polynomial $A(X) = \prod_{j=1}^{d-1} E_k(x_j^2; X) \in \mathbb{R}[X]$ has $\deg A \le (d-1)k < L := (d-1)k+1$. Real Lagrange interpolation (`Lagrange.eq_interpolate`) at the L real nodes $0, 1, \dots, L-1 \in \mathbb{R}$ expresses each $[X^q]A(X) = (\sum_{j=1}^{d-1} x_j^2)^q/q!$ $(0 \le q \le k)$ as an $\mathbb{R}$-linear combination of $\prod_{j=1}^{d-1} E_k(x_j^2; r)$, yielding $(d-1)k+1$ real d-way products.
- *Nonreal rates ($\mathrm{Im}\,\lambda > 0$, $P \in \mathbb{C}[X]$ of degree k):* We generalize Proposition 3.2 from scalars to polynomials of degree at most k ($d \ge 2$):

Proposition 3.4 (Universal root-of-unity polynomial-product identity, `re_coeff_prod_eq_sum_re`)**.** *Let $d \ge 2$, $k \ge 0$, and $R = 2(d-1)k + d$. For any $U_0, \dots, U_{d-1} \in \mathbb{C}[X]$ with $\deg U_j \le k$,*

$$\mathrm{Re}\left([X^k] \prod_{j=0}^{d-1} U_j(X)\right) = \frac{2^{d-1}}{R} \sum_{r=0}^{R-1} (-1)^r \prod_{j=0}^{d-1} \mathrm{Re}\Big(U_j\Big(e^{\frac{2\pi i r}{R}}\Big)\, e^{-\frac{i\pi r(2k+1)}{R}}\Big). \tag{7}$$

Proof. Expanding $\mathrm{Re}\, w = \frac{1}{2}(\bar{w}+w)$ over subsets $S \subseteq \{0,\dots,d-1\}$ of size $s = |S|$ and coefficient indices $q \in \{0,\dots,k\}^d$, and noting that $(-1)^r = e^{i\pi r R/R}$ with $R = (2k+1)d - 2k$, gives the r-th phase $\exp(2\pi i r K(S,q)/R)$ where

$$K(S,q) = (2k+1)s + \sum_{j \notin S} q_j - \sum_{j \in S} q_j - k \in \mathbb{Z}.$$

Because $0 \le q_j \le k$ and $d \ge 2$: (i) when $S = \emptyset$ ($s = 0$), $-R < -k \le K(\emptyset, q) = \sum_j q_j - k \le (d-1)k < R$, so $R \mid K(\emptyset, q) \iff \sum_j q_j = k$; (ii) when $S = \{0,\dots,d-1\}$ ($s = d$), $K = R + (k - \sum_j q_j)$, so $R \mid K \iff \sum_j q_j = k$; and (iii) when $1 \le s \le d-1$, $K(S,q) \ge (2k+1)s - sk - k = (k+1)s - k \ge 1$ and $K(S,q) \le (2k+1)s + (d-s)k - k = (k+1)s + (d-1)k \le 2(d-1)k + d - 1 = R - 1$, so $R \nmid K(S,q)$ and the geometric sum over $0 \le r < R$ vanishes identically. Applying (7) to $U_0(X) = e^{\lambda x_0^2} R(x_0^2; X)$ and $U_j(X) = e^{\lambda x_j^2} E_k(x_j^2; X)$ $(j \ge 1)$ yields $2(d-1)k + d$ real d-way products. □

Unified Alder–Strassen lower bound over $\mathbb{R}$ (`realExpForm_rank_ge`). Let $G = (\prod_{i\in\iota_{\mathbb{R}}} \mathbb{R}^{m_{\mathbb{R}}(i)}) \times (\prod_{k\in\iota_{\mathbb{C}}} \mathbb{C}^{m_{\mathbb{C}}(k)})$ with $m(i) = \deg P_i + 1$. Conjugate-rate uniqueness shows via duality that the exponential-monomial curve $s \mapsto ((s^q/q!\, e^{\mu_i s}), (s^q/q!\, e^{\lambda_k s}))$ spans G over $\mathbb{R}$ (`span_realExpEval_eq_top`), transferring any rank-R decomposition of the Hankel kernel to $\mathbb{R}$-linear functionals $\ell_{n,j} \in G^*$ for the combined truncated-convolution form `realExpForm`. Note first that every $L \in \mathrm{Hom}_{\mathbb{R}}(\mathbb{C}, \mathbb{R})$ has a nonzero kernel vector $w \in \ker L \setminus \{0\}$ (`exists_ker_ne_zero`), from which induction on $d \geq 2$ gives $d \leq R$ for any single nonzero complex product $\mathrm{Re}(\beta \prod_{j=0}^{d-1} z_j)$ (`single_complex_rank_ge`). Strong induction on the cutoff sum then peels:

1. *Real cutoffs (`realExpForm_rank_ge`):* Identical to Section 3.2 (1 term when $c_{\mathbb{R}}(i_0) = 1$, $d-1$ terms when $c_{\mathbb{R}}(i_0) \geq 2$), while keeping the $\mathbb{C}$-coordinates of G fixed so all nonreal blocks are untouched.
2. *Nonreal cutoff $c_{\mathbb{C}}(k_0) = 1$ (`cplxTruncForm_rank_ge`):* On mode 0, evaluating at 1_{k_0} and at $w_1 \in \ker(\ell_{n_1,0}|_{\mathbb{C}e_{k_0,0}}) \setminus \{0\}$ kills 2 distinct terms, and on each mode $j = 1, \ldots, d-2$, `single_complex_rank_ge` forces one additional active term outside the killed set D (else $|D| = j+1 < d$ terms would represent $\mathrm{Re}(\beta_{k_0} \prod z_q)$), reaching $|D| = d$ before zeroing block k_0 on mode $d-1$.
3. *Nonreal cutoff $c_{\mathbb{C}}(k_0) \geq 2$:* Let $q_{\mathrm{top}} = c_{\mathbb{C}}(k_0) - 1 \geq 1$ with weight $\beta_{\mathrm{top}} \neq 0$. In each mode $j = 0, \ldots, d-2$, evaluating at $u(z)$ with $\mathrm{emb}_{\mathrm{top}}(z)$ on mode j, $\mathrm{emb}_0(\overline{\beta_{\mathrm{top}} z})$ on mode $d-1$, and $\psi_m(\mathrm{emb}_0(1))$ elsewhere gives $|\beta_{\mathrm{top}} z|^2 > 0$ for every $z \in \mathbb{C} \setminus \{0\}$ (since $2q_{\mathrm{top}} \geq c_{\mathbb{C}}(k_0)$ kills all top-degree cross-terms). Testing $z = 1$ and $z = w_1 \in \ker(\ell_{n_1,j} \circ \mathrm{emb}_{\mathrm{top}}) \setminus \{0\}$ kills 2 distinct terms per mode—$2(d-1)$ terms across modes $0, \ldots, d-2$—after which low-degree truncation on modes $0, \ldots, d-2$ and $+1$ index shifting on mode $d-1$ reduce $c_{\mathbb{C}}(k_0) \mapsto c_{\mathbb{C}}(k_0) - 1$ with shifted top weight $\beta_{\mathrm{top}} \neq 0$.

Theorem 3.5 (Exact d-D Real Radial Rank, `separableRank_radial_eq_iff`)**.** *Let $d \geq 2$ and let $\varphi : \mathbb{R} \to \mathbb{R}$ be continuous on $[0, \infty)$. For every $N \in \mathbb{N}$,*

$$\mathrm{separableRank}\left(x \in \mathbb{R}^d \mapsto \varphi\left(\sum_{j=0}^{d-1} x_j^2\right)\right) = N$$

$$\iff \exists\, \mathrm{cs},\ \mathrm{IsRepresentation}(\mathrm{cs}) \wedge (\forall s \geq 0,\ \varphi(s) = \mathrm{profile}(\mathrm{cs}, s)) \wedge \mathrm{multiTotalCost}(d, \mathrm{cs}) = N.$$

In particular, for a Gaussian representation with r real and c nonreal rates, separableRank $= r + dc$ *(`separableRank_radial_gaussian_profile`), and in $d = 2$,* separableRank $=$ totalCost(cs) *(`separableRank_radial_two_eq_iff`).*

Theorem 3.6 (Minimal Differential Annihilator)**.** *Let $\varphi : \mathbb{R} \to \mathbb{R}$ be continuous on $[0, \infty)$ with finite-dimensional translate span and two-dimensional radial rank* separableRank$(x \in \mathbb{R}^2 \mapsto \varphi(x_0^2 + x_1^2)) = N < \infty$ *(`exists_minimal_differential_annihilator`). There exists a monic polynomial $p \in \mathbb{C}[X]$ of degree N such that for every $q \in \mathbb{C}[X]$, the half-line differential equation $q(D)\varphi_{\mathbb{C}} = 0$ holds on $[0, \infty)$ (where $\varphi_{\mathbb{C}}(s) = (\varphi(s) : \mathbb{C})$) if and only if $p \mid q$.*

Worked example ($d = 3, \deg P = 1$): exact rank vs. complex rank and border rank. Consider $K(x_0, x_1, x_2) = (x_0^2 + x_1^2 + x_2^2)\cos(x_0^2 + x_1^2 + x_2^2)$ on $\mathbb{R}^3$, a single well-formed component $P(s) = s$, $\lambda = i$ ($\deg P = 1, \mathrm{Im}\,\lambda = 1 > 0$):

- *Complex rank $= 6$:* Over $\mathbb{C}$, $K_{\mathbb{C}}$ splits into $\frac{1}{2}se^{is} + \frac{1}{2}se^{-is}$, giving exact complex 3-way rank $((3-1)\cdot 1 + 1) + ((3-1)\cdot 1 + 1) = 6$ by Theorem 3.1.
- *Pointwise limit of rank-6 kernels (border rank ≤ 6):* As an unformalized analytical contrast, for every $h > 0$ the difference quotient $\frac{1}{h}\left[\mathrm{Re}\prod_{j=0}^{2}(1 + hx_j^2)e^{ix_j^2} - \mathrm{Re}\prod_{j=0}^{2} e^{ix_j^2}\right]$ converges

pointwise (and uniformly on compact sets) to $K(x)$ as $h \to 0$, and by Proposition 3.2 each $\mathrm{Re}(\prod_{j=0}^{2} \cdots)$ uses 3 real 3-way products, totaling $3 + 3 = 6$ real products at every $h > 0$ with diverging $O(h^{-1})$ weights.

- *Exact real rank* $= 7$*:* By Theorem 3.5, $\mathrm{separableRank}_{\mathbb{R}}(K) = 2(3-1) \cdot 1 + 3 = 7$ (achieved with bounded weights by Proposition 3.4 at $R = 7$, vs. 9 from naive `IsSumOfProducts.re`). To see why 6 terms cannot represent K exactly, pass to $W = \mathbb{C}[T]/(T^2) \cong \mathbb{C} \oplus \mathbb{C}\varepsilon$ ($\varepsilon^2 = 0$) where $\mathcal{M}(u, v, w) = \mathrm{Re}(u_1 v_0 w_0 + u_0 v_1 w_0 + u_0 v_0 w_1)$. On $E = 0 \oplus \mathbb{C}\varepsilon \cong \mathbb{R}^2$, the slice $\mathcal{M}((0, z_1), (v_0, 0), (w_0, 0)) = \mathrm{Re}(z_1 v_0 w_0)$ forces the mode-0 functionals $\{\ell_n|_E\}$ to span E^* ($\dim_{\mathbb{R}} E^* = 2$); choosing a basis $\{\ell_0|_E, \ell_1|_E\}$ defines $\psi_0(u_0) = (u_0, c_0(u_0))$ killing $n = 0, 1$. Because $\varepsilon^2 = 0$, $\mathcal{M}(\psi_0(u_0), (0, v_1), (w_0, 0)) = \mathrm{Re}(u_0 v_1 w_0)$ is unchanged by c_0, so the remaining functionals $\{m_n|_E : n \geq 2\}$ on mode 1 again span E^*, killing two more terms $n = 2, 3$ via $\psi_1(v_0) = (v_0, c_1(v_0))$. Finally, $\mathcal{M}(\psi_0(u_0), \psi_1(v_0), (0, w_1)) = \mathrm{Re}(u_0 v_0 w_1)$ must be represented by the remaining $R - 4$ terms, and `single_complex_rank_ge` gives $R - 4 \geq 3$, hence $R \geq 7$.

4 Standard vs. Nonstandard Proofs of the Semigroup Step

Every proof of necessity above rests on the continuous one-parameter semigroup theorem: if $\mathcal{A}$ is a Banach algebra over $\mathbb{R}$ and $M : \mathbb{R} \to \mathcal{A}$ satisfies $M(0) = 1$, $M(s + t) = M(s)M(t)$ for all $s, t \geq 0$, and M is continuous on $[0, \infty)$, then there exists $A \in \mathcal{A}$ such that $M(t) = \exp(tA)$ for all $t \geq 0$.

4.1 Integral smoothing vs. Euler's infinitesimal step with homogeneous scale

Integral smoothing (`Semigroup.lean`). Since $M(0) = 1$ and M is continuous at 0, there exists $\delta > 0$ such that $I_\delta = \int_0^\delta M(u)\, du$ satisfies $\|I_\delta - \delta \cdot 1\| < \delta$, making I_δ invertible in $\mathcal{A}$ by Neumann series. For $t \geq 0$, changing variables in $M(t)I_\delta = \int_t^{t+\delta} M(u)\, du$ shows by the fundamental theorem of calculus that M is right-differentiable on $[0, \infty)$ with $M'(t) = M(t)A$, where $A = (M(\delta) - 1)I_\delta^{-1}$ commutes with $M(t)$. Differentiating $t \mapsto M(t)\exp(-tA)$ gives 0, so $M(t) = \exp(tA)$ in *any* Banach algebra.

Finite differences (`GeneratorBound.lean`, `FiniteDifferenceSemigroup.lean`). Without integration, if $\|M(u) - 1\| \leq \frac{1}{2}$ on $[0, \delta]$, geometric-series telescoping bounds the difference quotients $\|(M(\delta/n) - 1)/(\delta/n)\|$. In finite dimensions, compactness extracts a convergent subsequence $(M(h_k) - 1)/h_k \to A$; along h_k, the one-sided difference quotients of $M(t)\exp(-tA)$ tend to 0, and a one-sided mean-value lemma shows $M(t)\exp(-tA) = 1$.

Nonstandard analysis with homogeneous common scale. Euler's heuristic sets $A = \mathrm{st}((M(\varepsilon) - 1)/\varepsilon)$ for a positive infinitesimal $\varepsilon > 0$ and iterates $M(\varepsilon)^{\lfloor t/\varepsilon \rfloor} \approx (1 + \varepsilon A)^{\lfloor t/\varepsilon \rfloor} \approx \exp(tA)$. Directly applying st to $(M(\varepsilon) - 1)/\varepsilon$, however, requires proving beforehand that this quotient is finite (the 124-line geometric-sum bound above).

All three nonstandard formalizations in this project eliminate that *a priori* bound by passing to *projective / homogeneous coordinates*: put $x = M(\varepsilon) - 1$ (which is infinitesimal by continuity at 0) and define the common infinitesimal scale

$$\eta = \varepsilon + \|x\| > 0. \tag{8}$$

Because $\frac{\varepsilon}{\eta} + \left\|\frac{x}{\eta}\right\| = 1$ with $\frac{\varepsilon}{\eta} \in [0, 1]$ and $\left\|\frac{x}{\eta}\right\| \leq 1$, both rescaled quantities are bounded by 1 *without any analytical estimate.* In a finite-dimensional algebra $\mathcal{A}$, Heine–Borel compactness gives a standard vector $B = \mathrm{st}(x/\eta)$ with $\|B\| \leq 1$ and $\varepsilon/\eta \approx a := 1 - \|B\| \in [0, 1]$. Now

Table 1: Verified source sizes for the five complete proofs of the continuous one-parameter semigroup step $M(t) = \exp(tA)$. Every project helper lemma in each proof's module dependencies is counted in full (excluding standard Mathlib and Isabelle/HOL library imports).

Proof method	Prover / files	Phys.	Nonbl.	Scope
Integral smoothing (main)	Lean 4 (`Semigroup.lean`)	96	87	Any Banach algebra
Finite differences	Lean 4 (`GeneratorBound`, `FD`)	256	233	Finite-dim.
NSA (ultrafilter seq.)	Lean 4 (`NonstandardSemigroup`)	206	190	Finite-dim.
NSA (ultrapower API)	Lean 4 (`Ultrapower*` + shared)	399	344	Finite-dim.
NSA (star + `transfer`)	Isabelle (`Semigroup_NSA.thy`)	275	253	Finite-dim. ($\|1\| = 1$)

compare $N = \lfloor s/\eta \rfloor$ steps of $M(\varepsilon)$ (arriving at time $N\varepsilon \approx sa$) against N steps of $\exp(\eta B)$ (arriving at time $N\eta \approx s$) for an arbitrary standard $s \geq 0$:

$$\frac{\|M(\varepsilon) - \exp(\eta B)\|}{\eta} \leq \left\| \frac{x}{\eta} - B \right\| + \left\| B - \frac{\exp(\eta B) - 1}{\eta} \right\| \approx 0.$$

Summing N single-step errors of size $\eta \cdot o(1)$ with $N\eta \leq s$ yields $M(sa) = \exp(sB)$ for all standard $s \geq 0$. Could the time coordinate vanish ($a = 0$)? If $a = 0$, then $\exp(sB) = M(0) = 1$ for every $s \geq 0$; differentiating at $s = 0$ gives $B = 0$, contradicting $a + \|B\| = 1$! Hence $a > 0$, and setting $A = B/a$ with $s = t/a$ gives $M(t) = \exp(tA)$.

4.2 Three nonstandard formalizations and quantitative comparison

We formalized the homogeneous-scale nonstandard proof in three settings:

1. **Isabelle/HOL:** `Isabelle/Semigroup_NSA.thy` uses `HOL-Nonstandard_Analysis`, which provides hyperreals, hypernaturals, star extensions `*f*`, and the `transfer` proof method. Single-step error accumulation reduces to a purely algebraic noncommutative power estimate (`power_difference_bound`): if $\|u\|, \|v\| \leq e^h$ with $h \geq 0$, then $\|u^n - v^n\| \leq ne^{nh}\|u - v\|$, proved by induction from $u^{n+1} - v^{n+1} = u(u^n - v^n) + (u - v)v^n$. Since $\|M(\varepsilon)\| \leq 1 + \|x\| \leq 1 + \eta \leq e^\eta$ and $\|\exp(\eta B)\| \leq e^{\eta\|B\|} \leq e^\eta$, `transfer` applies this ordinary inequality directly to a hypernatural step count N. (This uses $\|1\| = 1$ from Isabelle's `real_normed_algebra_1`; in Lean, $\mathcal{L}(\mathcal{W}(\psi))$ has $\|1\| = 0$ when $\psi = 0$, so the Lean theorems avoid assuming $\|1\| = 1$ and telescope steps using compact-interval bounds via `norm_replace_steps_le` instead.)

2. **Lean 4 with ultrafilter sequences:** `Experiments/NonstandardSemigroup.lean` expresses "infinitely close" directly as convergence along hyperfilter $\mathbb{N}$ (since Mathlib defines Hyperreal = Germ(hyperfilter $\mathbb{N}$) $\mathbb{R}$ without a transfer tactic), obtaining B in one line via `IsCompact.ultrafilter_le_nhds`.

3. **Lean 4 in nonstandard vocabulary:** In `Experiments/`, `Ultrapower.lean` and `UltrapowerSemigroup.lean` build a reusable 195-line ultrapower interface (`Hyp`, `std`, `ext`, $\approx_{\mathrm{h}}$, one-variable transfer lemmas) and write Euler's proof in classical nonstandard notation.

Table 1 summarizes the exact line counts. These numbers reflect *library amortization* rather than intrinsic proof complexity: `Semigroup.lean` (96 lines) is shortest and holds in every Banach algebra because Mathlib amortizes thousands of lines of Bochner integration, Neumann series, and the fundamental theorem of calculus behind compact API lemmas, whereas compactness-based proofs are inherently finite-dimensional. Across different proof styles within Lean, the 2× gap between integral smoothing (96 lines) and ultrafilter sequences (206 lines) remained stable even after eliminating the 124-line difference-quotient bound via homogeneous

coordinates (8). Moreover, discounting the 195-line interface `Ultrapower.lean` as reusable library code leaves $169+35=204$ lines: without automated transfer, nonstandard vocabulary in Lean merely wraps filter limits. In Isabelle/HOL, `transfer` genuinely compresses hypernatural induction, but 59 lines of local library gaps (Heine–Borel vector standard parts and half-line continuity via $M(\max(t,0))$) plus stating transferred formulas at both standard and nonstandard types keep the total at 275 lines.

5 Applications to 2-D Blur, Steerable Filtering, and 3-D Acoustic Propagation

All numerical optimizations, finite-precision simulations, and GPU benchmarks in Sections 5.2–5.3 are implemented in reproducible scripts under `Applications/` and executed on an NVIDIA GeForce RTX 4070 Ti Super (16 GiB GDDR6X, 672 GB/s peak memory bandwidth). Before turning to specific designs, note an analytical duality linking spatial and frequency-domain radial kernels: for $\operatorname{Re}\lambda<0$, differentiating $\mathcal{F}_{\mathbb{R}^d}[e^{\lambda\|x\|^2}](\xi)=(\pi/(-\lambda))^{d/2}\exp(\|\xi\|^2/(4\lambda))$ k times with respect to λ maps $P(\|x\|^2)e^{\lambda\|x\|^2}$ ($\deg P=k$) to $\widetilde{P}(\|\xi\|^2)e^{\|\xi\|^2/(4\lambda)}$ with $\deg\widetilde{P}=k$ and $\operatorname{Re}(1/(4\lambda))<0$. By Theorems 3.1 and 3.5, the d-dimensional Fourier transform preserves exact complex and real d-way separable rank across all integrable radial exponential-polynomial kernels.

5.1 Harmonic "beach-ball" differential operators and steerable kernels

Specializing Proposition 3.2 to equal symbolic factors $z_0=\cdots=z_{d-1}=\partial_x+i\partial_y$ (writing m for the harmonic order and $D_\alpha=\cos\alpha\,\partial_x+\sin\alpha\,\partial_y$ for the directional derivative along angle α) gives, for *every* integer $m\geq 1$ (both odd and even),

$$\operatorname{Re}\big((\partial_x+i\partial_y)^m\big)=\frac{2^{m-1}}{m}\sum_{k=0}^{m-1}(-1)^k D_{\pi k/m}^m. \tag{9}$$

On a regular $2m$-point ring $\theta_n=\pi n/m$ ($n=0,\dots,2m-1$) of radius $h\to 0$, antipodal points satisfy $D_{\theta_{k+m}}=-D_{\theta_k}$ while Fourier orthogonality of $(-1)^k=e^{i\pi k}$ annihilates all Taylor orders below m. For $f\in C^m$ near (x,y), (9) recovers—by elementary geometric series rather than Cauchy residues and Chebyshev partial fractions—both the odd-order ($m=2N+1$) hexagonal/decagonal "beach-ball" operators of Niemitalo [19] and the even-order operators of Perona [22] in a single formula:

$$\lim_{h\to 0}\frac{1}{h^m}\sum_{n=0}^{2m-1}(-1)^n f\Big(x+h\cos\frac{\pi n}{m},\,y+h\sin\frac{\pi n}{m}\Big)=\frac{1}{(m-1)!\,2^{m-2}}\operatorname{Re}\big((\partial_x+i\partial_y)^m\big)f(x,y),$$

with a half-sector ring rotation $+\frac{\pi}{2m}$ yielding $+\operatorname{Im}((\partial_x+i\partial_y)^m)$. Recognizing $\partial_{\bar z}=\frac{1}{2}(\partial_x+i\partial_y)$ as the Cauchy–Riemann operator also explains why applying $\partial_{\bar z}^m$ to *any* smooth radial kernel $\varphi(x^2+y^2)=\varphi(z\bar z)$ produces no lower-order radial derivatives: because $\partial_{\bar z}(z)=0$, induction gives

$$\left(\frac{\partial_x+i\partial_y}{2}\right)^m\varphi(x^2+y^2)=(x+iy)^m\,\varphi^{(m)}(x^2+y^2)=r^m e^{im\theta}\,\varphi^{(m)}(r^2). \tag{10}$$

When $\varphi(s)$ is a well-formed multi-Gaussian profile with r real and c nonreal upper-half-plane rates (all $\lambda_i\neq 0$, approximating, e.g., an ideal circular low-pass filter [20]), $\varphi^{(m)}(s)$ is *again* a well-formed multi-Gaussian profile at the same rates (weighted by λ_i^m) of continuous 2-way rank $R=r+2c$. Analytically differentiating the continuous-space kernel on $\mathbb{R}^2$—avoiding discrete finite-difference distortion up to Nyquist [18]—absorbs the monomials of $\operatorname{Re}((x+iy)^m)$

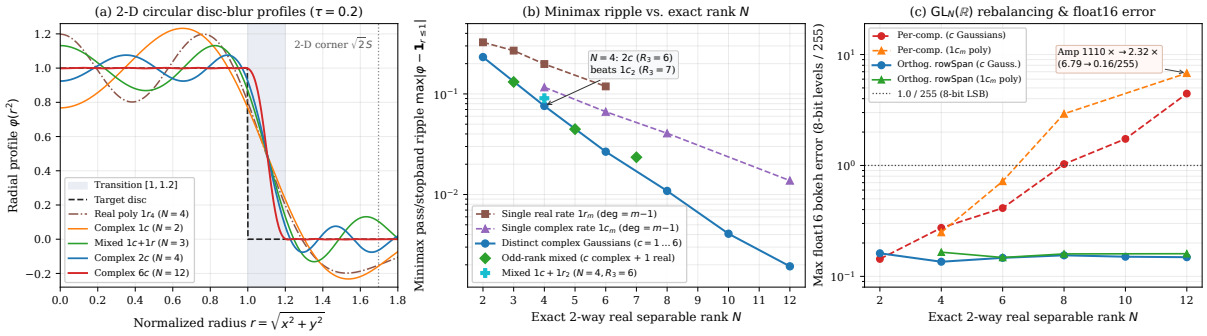


Figure 1: Two-dimensional circular disc blur ($\tau = 0.2$, `Applications/blur_2d.py`): (a) minimax radial profiles $\varphi(r^2)$ against the unit disc; (b) minimax pass/stopband ripple across exact 2-way real separable ranks $N \in [2, 12]$ for distinct complex Gaussians, odd-rank mixed Gaussians, and repeated-rate polynomial-Gaussians; (c) maximum IEEE `float16` bokeh rendering error on a 512×512 HDR scene (peak 8.0, radius $R_{\text{px}} = 25$, tap width $W = 61$) before and after nuclear-norm-optimal $\mathrm{GL}_N(\mathbb{R})$ rebalancing on rowSpan(K).

and $\mathrm{Im}((x+iy)^m)$ into the 1-D Gaussian factors, giving exact *continuous-domain* 2-way real separable ranks on $\mathbb{R}^2$:

$$\big(\lfloor m/2 \rfloor + 1\big)(r + 2c) \quad \text{for } \cos(m\theta), \qquad \lceil m/2 \rceil (r + 2c) \quad \text{for } \sin(m\theta).$$

Sampling those 1-D factors on a finite grid yields a discrete separable realization with at most that many terms (grid aliasing at special rates such as $\mathrm{Im}\,\lambda_i \in \pi\mathbb{Z}$ can collapse the sampled discrete rank further). On the 161×161 ($\omega_c = \pi/64$) orientation-estimation filter of [18], replacing direct $2 \times 161^2 = 51{,}842$ taps/pixel by a two-complex-Gaussian ($R = 4$) analytical gradient pair reduces scalar multiply–adds to 2,576 (20×, or 40× with even/odd tap-parity folding).

Continuous-domain pass-count bounds for simultaneous steerable pairs. Can sharing intermediate 1-D continuous convolutions reduce the simultaneous evaluation of $(C_m * I, S_m * I)$ below separate evaluation $2(\lfloor m/2 \rfloor + 1)R + 2\lceil m/2 \rceil R = 2(m+1)R$? No: if any two-stage 1-D schedule on $\mathbb{R}^2$ splits $(C_m, S_m) = (A_c + B_c, A_s + B_s)$ into P_y shared vertical passes followed by $\mathrm{rank}(A_c) + \mathrm{rank}(A_s)$ horizontal passes plus P_x shared horizontal passes followed by $\mathrm{rank}(B_c) + \mathrm{rank}(B_s)$ vertical passes, then

$$\begin{aligned} \dim \mathrm{rowSpan}_x(C_m, S_m) &\le \mathrm{rank}(A_c) + \mathrm{rank}(A_s) + P_x, \\ \dim \mathrm{colSpan}_y(C_m, S_m) &\le P_y + \mathrm{rank}(B_c) + \mathrm{rank}(B_s). \end{aligned}$$

Because C_m and S_m on $\mathbb{R}^2$ span all powers $\{x^0, \dots, x^m\} g_n(x)$ in x and $\{y^0, \dots, y^m\} g_n(y)$ in y, adding the two inequalities shows that *every* two-stage continuous 1-D schedule requires at least $2(m+1)R$ passes ($4R$ passes for (h_x, h_y)). Savings instead come from: (i) definite even/odd parity of every 1-D factor $u^k g_n(u)$, halving multiplications per sampled pass via $I[y+\ell] \pm I[y-\ell]$ folding; (ii) sharing $\{I *_y (y^k g_n)\}_{k=0}^{M}$ across *multiple* harmonic orders ($5R$ instead of $6R$ for $m = 0, 1$; $10R$ instead of $12R$ for $m = 1, 3$); and (iii) for the d-D gradient $(h_{x_0}, \dots, h_{x_{d-1}})$ of a multi-Gaussian radial kernel ($d \ge 3$), recursively splitting $\{0, \dots, d-1\}$ into halves to reduce $d^2 R_d$ passes to $T(d) R_d$ with $T(1) = 1$ and $T(d) = T(\lfloor d/2 \rfloor) + T(\lceil d/2 \rceil) + d \le d(\lceil \log_2 d \rceil + 1)$ ($8R_3$ vs. $9R_3$ in $d = 3$; $12R_4$ vs. $16R_4$ in $d = 4$).

5.2 Worked application I: 2-D circular lens blur and $\mathrm{GL}_N(\mathbb{R})$ conditioning

In depth-of-field bokeh rendering [17, 21], the ideal defocus point-spread function is the discontinuous unit disc $\mathbf{1}_{r \le 1}$ ($r = \sqrt{x^2 + y^2}$), which has infinite separable rank by Theorem 3.3.

Table 2: Representative 2-D circular disc-blur designs ($\tau = 0.2$, $R_{\text{px}} = 25$, $W = 61$). Exact 2-D rank N and 3-D ball rank R_3 follow Theorem 3.5; SVD is the empirical singular-value count ($\sigma_k > 10^{-10}\sigma_1$) of the 61×61 kernel; $\kappa_\infty = (\sum_n \|h_n\|_\infty \|v_n\|_\infty)/\|K\|_\infty$ measures peak channel amplification before (raw) and after (orth) rowSpan rebalancing, alongside max `float16` HDR bokeh error (in 8-bit units /255).

Structure	Degrees	N (d=2)	SVD	R_3 (d=3)	Ripple $\pm\delta$	$\kappa_\infty^{\text{raw}}$	$\kappa_\infty^{\text{orth}}$	`fp16` raw	`fp16` orth
$1c$ (Gaussian)	0	2	2	3	0.2325	1.65	1.30	0.14	0.16
$1c + 1r$ (mixed)	0, 0	3	3	4	0.1315	3.45	1.65	0.15	0.15
$2c$ (Gaussians)	0, 0	4	4	6	0.0759	5.28	1.95	0.27	0.14
$1c + 1r_2$ (mixed poly)	0, 1	4	4	6	0.0905	4.50	1.92	0.20	0.15
$1c_2$ (complex poly)	1	4	4	7	0.1165	5.68	1.88	0.25	0.17
$1r_4$ (real Laguerre)	3	4	4	7	0.1982	4.74	1.56	0.31	0.15
$2c + 1r$ (mixed)	0, 0, 0	5	5	7	0.0446	8.73	2.08	0.25	0.13
$4c$ (Gaussians)	0^4	8	8	12	0.0109	70.06	2.31	1.03	0.15
$6c$ (Gaussians)	0^6	12	12	18	0.0019	595.01	2.49	4.44	0.15
$1c_6$ (complex poly)	5	12	12	23	0.0138	1,109.75	2.32	6.79	0.16

Introducing a transition band $[1, S]$ with $S = 1 + \tau$ ($\tau = 0.2$) and truncating each 1-D pass to $|x|, |y| \le S$ restricts the 2-D separable support $[-S, S]^2$ to radii $r \le \sqrt{2}\,S \approx 1.697$. For any fixed choice of decay and oscillation rates (b_u, c_u), minimizing the Chebyshev ripple $\max_{r \in [0,1] \cup [S, \sqrt{2}S]} |\varphi(r^2) - \mathbf{1}_{r \le 1}|$ over all polynomial weights is a linear program (solved with HiGHS inside differential-evolution/Nelder–Mead rate search in `Applications/blur_2d.py`).

Figure 1 and Table 2 highlight three structural consequences of Theorem 3.5:

1. **Role of nonreal rates, odd budgets, and 3-D lifting:** Across all 18 tested families, empirical SVD rank matches c.multiCost(2) identically. Distinct complex Gaussians achieve geometric ripple decay with $N = 2c$ ($0.2325 \to 0.0019$ from $1c$ to $6c$, or 0.0014 under global Remez search [21]), while adding a single real Gaussian ($1c+1r, 2c+1r$) interpolates cleanly at odd ranks $N = 3, 5$. Free real-only rates coalesce into repeated-rate Laguerre–Gaussians ($1r_m$). At $N = 4$, four distinct structures tie in 2-D cost, but lifting to a 3-D spherical ball splits their exact 3-way ranks: $2c$ gives both the lowest ripple (± 0.0759) and lower 3-D rank ($R_3 = 6$ vs. $R_3 = 7$ for $1c_2$ and $1r_4$).

2. **Basis non-uniqueness in $d = 2$ vs. Kruskal uniqueness in $d \ge 3$:** While $N = \dim \text{rowSpan}(K)$ is an algebraic invariant, any $T \in \text{GL}_N(\mathbb{R})$ maps $(h, v) \mapsto (Th, T^{-\top}v)$ to another rank-N realization. Splitting high-order approximations component-by-component causes severe intermediate cancellation ($\kappa_\infty^{\text{raw}} = 595$ on $6c$ and 1,110 on $1c_6$), producing visible 4.44/255 and 6.79/255 errors around HDR specular highlights when intermediate passes are stored in IEEE `float16` (as in WebGPU `pack2x16float` [21]). Orthonormalizing the 1-D basis via QR and symmetric eigendecomposition $K(x, y) = \sum_{n=1}^{N} e_n g_n(x) g_n(y)$ ($\|g_n\|_2 = 1$, $h_n = \sqrt{|e_n|}\,g_n$, $v_n = \text{sign}(e_n) h_n$, accumulated in ascending order of $|e_n|$) achieves the global nuclear-norm minimum $\sum_n \|h_n\|_2 \|v_n\|_2 = \sum_n |e_n| = \|K\|_{S_1}$, collapsing κ_∞ to ≤ 2.49 and keeping `float16` max error at 0.13–0.17/255 (RMSE ≤ 0.033/255) across all $N \le 12$. By contrast, in $d \ge 3$, Kruskal's theorem [11] implies that minimal d-way CP decompositions have *no* continuous $\text{GL}_R(\mathbb{R})$ rotation freedom (only permutation and scaling), making well-separated rates and amplitude-balanced root-of-unity factors essential.

3. **GPU throughput:** Using two-pass symmetric Triton kernels with constant two-buffer memory on 1920×1080 RGBA images at radius $R_{\text{px}} = 64$ ($W = 155$), direct 155×155 depthwise `conv2d` takes 137.1 ms, whereas rank $N = 4, 8, 12$ takes 7.01, 14.17, 21.19 ms in `float32` ($19.6\times, 9.7\times, 6.5\times$) and 3.99, 8.05, 12.09 ms in `float16` ($34.3\times, 17.0\times, 11.3\times$).

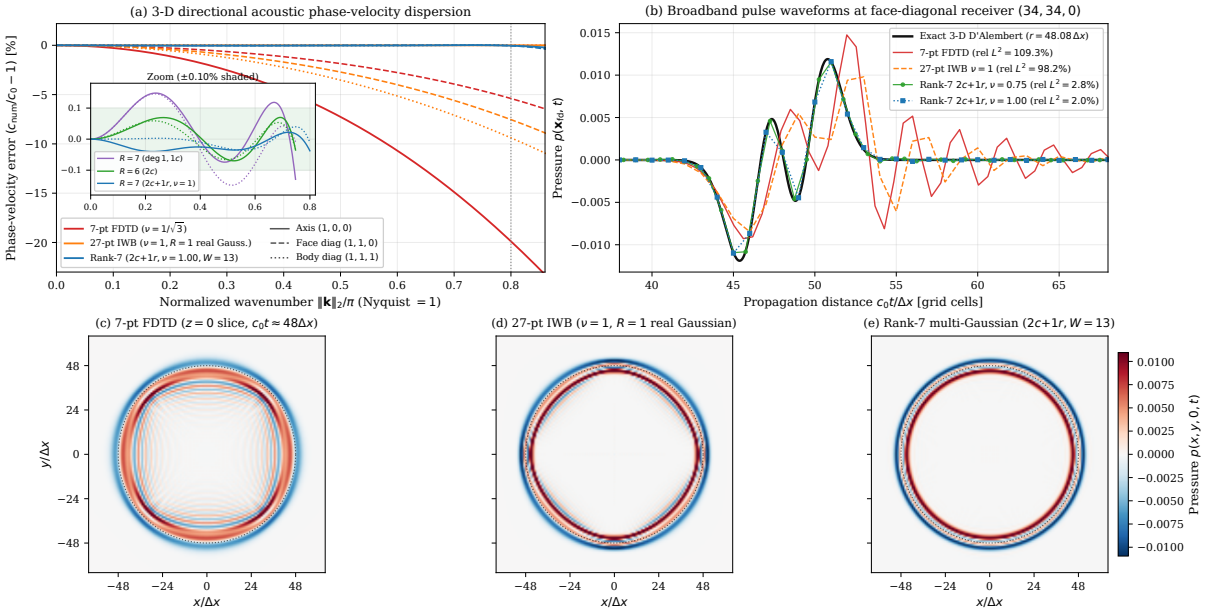


Figure 2: Three-dimensional acoustic wave propagation (`Applications/acoustics_3d.py`): (a) directional phase-velocity error ($c_{\text{num}}(\mathbf{k})/c_0 - 1$) along $(1,0,0)$, $(1,1,0)$, and $(1,1,1)$ across $\|\mathbf{k}\|_2/\pi \in [0, 0.86]$, with inset zoom ($\pm 0.10\%$ shaded) for $R = 6, 7$; (b) broadband pulse waveforms ($\sigma_0 = 1.40\,\Delta x$) on a 256^3 grid at the face-diagonal receiver $(34, 34, 0)$ ($r = 48.08\,\Delta x$) vs. the exact 3-D D'Alembert solution; (c–e) mid-plane wavefield slices $p(x, y, 0, t)$ at $c_0 t \approx 48\,\Delta x$ (dotted circle: $r = 48\,\Delta x$).

5.3 Worked application II: isotropic 3-D acoustic wave simulation

Consider the 3-D constant-speed scalar acoustic wave equation $\partial_{tt} p = c_0^2 \nabla^2 p$ discretized on $\mathbb{Z}^3$ with spacing Δx and time step $\Delta t = \nu \Delta x / c_0$ (Courant number $\nu > 0$) by the explicit two-step leapfrog update

$$p^{n+1}[\mathbf{m}] + p^{n-1}[\mathbf{m}] = (M * p^n)[\mathbf{m}], \qquad \mathbf{m} \in \mathbb{Z}^3. \tag{11}$$

In discrete wavenumber $\mathbf{k} = (k_x, k_y, k_z) \in [-\pi, \pi]^3$, the exact continuous wave-equation propagator has radial symbol $\widehat{M}_{\text{exact}}(\mathbf{k}) = 2\cos(\nu \|\mathbf{k}\|_2)$, and (11) is von Neumann stable and energy-conserving on $\ell^2(\mathbb{Z}^3)$ if and only if $\widehat{M}(\mathbf{k}) \in [-2, 2]$ for all $\mathbf{k} \in [-\pi, \pi]^3$ (both characteristic roots of $\zeta^2 - \widehat{M}(\mathbf{k})\zeta + 1 = 0$ then lie on $|\zeta| = 1$). Standard 7-point FDTD ($\nu \le 1/\sqrt{3}$) suffers up to 19.7% axial phase-velocity error by $\|\mathbf{k}\|_2 = 0.80\pi$. Two structural facts explain why radial multi-Gaussians are natural here:

- **Inadequacy of additive axial stencils of any order:** Any purely axial scheme $\widehat{M}_{\text{ax}}(\mathbf{k}) = 2 + A(k_x) + A(k_y) + A(k_z)$ ($A(0) = 0$) lacks mixed cross-terms $k_x^2 k_y^2$: even if $A(k) = 2(\cos(\nu k) - 1)$ has zero dispersion along the coordinate axes, along $(1,1,1)/\sqrt{3}$ it gives $2 + 6(\cos(\nu k/\sqrt{3}) - 1) = 2 - \nu^2 k^2 + \frac{1}{36}\nu^4 k^4 - \ldots \neq 2\cos(\nu k) = 2 - \nu^2 k^2 + \frac{1}{12}\nu^4 k^4 - \ldots$, leaving an irreducible $O(\nu^4 k^4)$ anisotropy gap while Brillouin-corner stability $\widehat{M}_{\text{ax}}(\pi, \pi, \pi) \ge -2$ forces $A(\pi) \ge -4/3$ ($\nu k_c \le \arccos(1/3) \approx 0.392\pi$ when A tracks $2(\cos(\nu k) - 1)$).
- **Kowalczyk–van Walstijn IWB is a single real Gaussian:** In room acoustics, Kowalczyk and van Walstijn's 27-point Interpolated Wideband (`IWB`) stencil [10] ($a = \frac{1}{4}, b = \frac{1}{16}$, $\nu \le 1$) has face, edge, and corner weights $\frac{1}{4}\nu^2$, $\frac{1}{8}\nu^2$, $\frac{1}{16}\nu^2$ and center weight $2 - \frac{7}{2}\nu^2$, which is *identically*

$$M_{\text{IWB}}[\mathbf{m}] = (2 - 4\nu^2)\delta_{\mathbf{0}}[\mathbf{m}] + \frac{\nu^2}{2}\, 2^{-(m_x^2 + m_y^2 + m_z^2)} \qquad (\mathbf{m} \in \{-1, 0, 1\}^3),$$

Table 3: Certified 3-D acoustic propagators vs. FDTD baselines on 256^3 voxels (RTX 4070 Ti Super). Here $|\delta c/c_0|$ is the worst-case phase-velocity error up to k_c (0.80π for baselines); $k_{0.1\%}$ and $k_{0.5\%}$ are the largest isotropic cutoffs (in units of π) keeping $|\delta c/c_0| \le 0.1\%$ and 0.5% over all 3-D angles; Wave L^2 is the worst-case receiver relative L^2 waveform error at $r \approx 48\,\Delta x$; GPU step times compare direct W^3 `conv3d`, naive $(r+4c)$/9-term Triton passes, and exact R-term Triton passes in `fp32` and `fp16`-intermediate precision.

Scheme	ν	W	R (nv)	$\lvert\delta c/c_0\rvert$	$k_{0.1\%}$	$k_{0.5\%}$	Wave L^2	Direct	Naive	Exact `fp32`	Exact `fp16`
7-pt FDTD	0.58	3	—	19.71%	0.062π	0.141π	109.3%	—	—	0.323 ms	—
13-pt $O(\Delta x^4, \Delta t^2)$	0.50	5	—	8.59%	0.100π	0.224π	—	—	—	—	—
27-pt IWB $(1r)$	1.00	3	1 (1)	9.23%	0.091π	0.199π	98.2%	—	—	0.323 ms	—
Rank-4 $(1c{+}1r, 0.75\pi)$	0.75	9	4 (5)	0.316%	0.14π	0.77π	11.9%	76.8 ms	4.04 ms	3.30 ms	2.39 ms
Rank-6 $(2c, 0.75\pi)$	0.75	11	6 (8)	0.070%	0.76π	0.81π	5.2%	139.3 ms	6.29 ms	4.80 ms	3.44 ms
Rank-7 $(\deg 1\,1c, 0.75\pi)$	0.75	13	7 (9)	0.148%	0.15π	0.80π	6.7%	230.5 ms	7.09 ms	5.59 ms	4.02 ms
Rank-7 $(2c{+}1r, 0.80\pi)$	0.75	13	7 (9)	0.082%	0.81π	0.84π	4.0%	230.5 ms	7.09 ms	5.59 ms	4.02 ms
Rank-7 $(2c{+}1r, 0.80\pi)$	1.00	13	7 (9)	0.055%	0.82π	0.87π	2.4%	230.5 ms	7.09 ms	5.59 ms	4.02 ms
Rank-9 $(3c, 0.85\pi)$	0.75	17	9 (12)	0.061%	0.86π	0.89π	2.7%	104.7* ms	10.01 ms	7.66 ms	5.43 ms

the $(r = 1, c = 0, L = 1, R = 1)$ single-real-Gaussian member of our radial family (plus a zero-pass center tap). Because a single positive real Gaussian cannot match higher radial derivatives of $2\cos(\nu\|\mathbf{k}\|_2)$, `IWB` at $\nu = 1$ is exact on axes but incurs 9.23% diagonal phase-velocity error by 0.80π (Figure 2a,d).

Certified Brillouin-octant minimax design and exact Neumann walls. Generalizing `IWB` to wider stencils $\mathbf{m} \in [-L, L]^3$, we set $M[\mathbf{m}] = \alpha_0\delta_{\mathbf{0}}[\mathbf{m}] + \varphi(m_x^2 + m_y^2 + m_z^2)$, where α_0 fixes $\widehat{M}(\mathbf{0}) = 2$ identically and φ has exact 3-way real rank $R = r + 3c$ (Proposition 3.2) or $R = 7$ for $\deg P = 1$ (Proposition 3.4; verified to $\le 5.6 \times 10^{-16}$ on $[-L, L]^3$). For fixed rates, minimizing $\max_{0<\|\mathbf{k}\|_2\le k_c} |\widehat{M}(\mathbf{k}) - 2\cos(\nu\|\mathbf{k}\|_2)|/(2\nu\|\mathbf{k}\|_2 \sin(\nu\|\mathbf{k}\|_2))$ subject to exact asymptotic sound speed $\partial_{k_x}^2 \widehat{M}(\mathbf{0}) = -2\nu^2$ and strict von Neumann stability $-2 + 10^{-3} \le \widehat{M}(\mathbf{k}) \le 2 - \min(10^{-3}, \frac{1}{4}\nu^2\|\mathbf{k}\|_2^2)$ over all 47,904 symmetry-inequivalent nonzero points of a 65^3 grid on $[0, \pi]^3$ is solved via cutting-plane HiGHS linear programming. Moreover, because every 1-D radial factor $h[m] = h[-m]$ is even, half-sample symmetric index reflection at domain boundaries simultaneously diagonalizes all $3R$ 1-D passes in the 3-D `DCT-II` cosine basis—enforcing exact rigid-wall Neumann boundary conditions ($\partial_n p = 0$) without padding buffers or boundary instabilities. Across 1,000 steps (> 18 wall reflections) in a 64^3 rigid enclosure, discrete leapfrog energy drifted by at most 1.1×10^{-6} (`float32` rounding).

Table 3 and Figure 2 summarize accuracy and GPU performance (*at $W = 17$, cuDNN switches `conv3d` to an FFT algorithm, which the local $R = 9$ Triton cascade still outperforms by 13.7×–19.3×). Both dedicated 1-pass baselines (7-pt FDTD and 27-pt IWB) run at 0.323 ms/step (51.9 GVox/s, 92% of peak GDDR6X memory bandwidth). On the 256^3 broadband spherical pulse test ($\sigma_0 = 1.40\,\Delta x$), 7-pt FDTD and 27-pt IWB break into anisotropic dispersive wave trains (109.3% and 98.2% waveform error), whereas `Rank-7` $(2c + 1r, \nu = 1.00)$ keeps $|\delta c/c_0| \le 0.055\%$ (< 1 cent) up to 0.80π (2.44 points per wavelength at $k_{0.1\%} = 0.82\pi$), achieving 2.0%–2.4% broadband waveform error across all axes and diagonals (and 4.1% vs. 4.0% with `float16` intermediates at $\nu = 0.75$, since the LP weights satisfy $\|w\|_1 \le 1.51$ without cancellation). Replacing naive $(r + 4c) = 9$ expansion by Proposition 3.2's exact $R = 7$ formula saves 21.2% wall-clock time (5.59 ms vs. 7.09 ms, 41.3× faster than direct 13^3). Most importantly, because 3-D grid cost scales as $(\Delta x)^{-3}(\Delta t)^{-1} \propto k_{\text{tol}}^{-4}$, matching `Rank-7`'s 0.1%-accurate bandwidth (0.82π vs. 0.091π for 27-pt IWB at $\nu = 1$) requires a 9.01× finer grid per axis in 27-pt IWB—consuming 731× more GPU memory and 6,590× more voxel-steps, making `Rank-7` 381× faster in `float32` (529× in `fp16`-intermediate) than hardware-roofline 27-pt IWB at equal dispersion accuracy (21×–29× faster and 83× leaner even at a 0.5% tolerance).

Table 4: Structure and source line counts of the `KernelRankLean` library.

Module(s)	Mathematical content	Phys.	Nonbl.
`Defs`, `KernelRankLean`	Multi-way rank, surjections, slices, costs	213	180
`RowSpan`, `Complexification`	2-way row-span rank; $\mathbb{R} \to \mathbb{C}$ invariance	221	189
`Translations`	d-D radial-to-Hankel reduction; `translateSpan`	92	80
`ExpPoly`, `RealExpPoly`	Exponential-polynomial uniqueness	198	170
`Semigroup`, `TranslationSemigroup`	Banach semigroups; shift semigroup	262	219
`Necessity`, `Representation`	Generalized eigenvectors; conjugate pairing	222	198
`Differential`, `ComplexRank`	Half-line operators $q(D)$; 2-D complex rank	318	284
`Annihilator`	Minimal constant-coefficient annihilator	69	62
`MultiComplexRank`	Exact d-way complex rank $\sum_i((d-1)\deg P_i + 1)$	1,015	978
`MultiRealRank`	Exact d-way real rank (`multiTotalCost`)	2,562	2,521
`Main`	Unified d-D and 2-D main theorems	152	140
Total (`KernelRankLean`)		**5,324**	**5,021**

6 Formalization Architecture and Verification Audit

Table 4 breaks down the `KernelRankLean` library (5,324 physical lines, 5,021 nonblank lines; earlier stages comprised 3,822 physical lines at commit `6aabb4d` for d-D complex and multi-Gaussian ranks, 1,587 lines at commit `cae13d6` for 2-D plus differential annihilators, and 1,340 lines at commit `240dea0` for 2-D classification alone). Three formalization choices proved decisive in Lean 4:

1. **Dependent multi-way definitions with homogeneous two-coordinate specialization:** Defining `IsSumOfProducts` and `separableRank` over a general dependent family $\prod_{i\in\iota} X_i$ unifies 2-D and d-D statements, whereas specializing `rowSpan` in `RowSpan.lean` to homogeneous $\operatorname{Fin} 2 \to \alpha$ (with a pattern-matched constructor `fin2Vec`) avoids expensive higher-order unification timeouts when rewriting on $\operatorname{Fin} 2 \to \mathbb{R}_{\geq 0}$.
2. **Explicit uniformity selection on `translateSpan`:** Transferring a norm from $\mathbb{C}^n$ onto the finite-dimensional subspace $\mathcal{W}(\psi) \leq (\mathbb{R}_{\geq 0} \to \mathbb{C})$ creates a second `UniformSpace` instance alongside the subtype uniformity; explicitly installing the norm-induced uniformity as the local instance prevents typeclass diamond timeouts while letting `LinearMap.continuous_of_finiteDimensional` supply automatic continuity.
3. **Cayley–Hamilton divisibility in place of cyclic vectors:** Bounding $\dim_{\mathbb{C}} \mathcal{W}(\psi)$ from below via divisibility of $\chi_A(X)$ by $\prod_\mu (X-\mu)^{\deg P_\mu + 1}$ eliminated both coefficient-synthesis operators and cyclic-vector constructions, shrinking the 2-D proof plus differential-annihilator extension by 93 lines while yielding Theorem 3.6 directly.

Verification audit. A single script (`scripts/check`, also wired as the repository's pre-commit hook) builds both `KernelRankLean` and `Experiments` via `lake build` (which includes every `.lean` module in the repository), checks the compiler log to ensure no declaration uses `sorry`, runs `#print axioms` on 17 top-level theorems to confirm dependence solely on Lean's three standard classical axioms (`propext`, `Classical.choice`, `Quot.sound`), builds the Isabelle2026 session `KernelRank_NSA`, and audits `Semigroup_NSA.semigroup_eq_exp` via `Thm_Deps.all_oracles` (catching `skip_proof`/`sorry` even under `quick_and_dirty`) while rejecting any `axiomatization` or `axioms` declarations.

Acknowledgments

This work builds directly on problems, mathematical formulations, and software developed by Olli Niemitalo, including `kernel-optimizer` [21] for complex-Gaussian approximations of 2-D circularly symmetric blur kernels and his investigations of isotropic low-pass filters, steerable gradient orientation estimation, and planar harmonic differential operators [18, 19, 20]. Olli Niemitalo initiated and guided the `kernel-rank-lean` repository, proposed comparing standard and Isabelle/HOL nonstandard proofs of the semigroup step, and suggested extending the formalization to d dimensions. We also thank the aiXiv Official Agent reviewer on version 1.0, whose question regarding higher-degree real polynomial profiles in $d \geq 3$ directly motivated Theorem 3.5 and Proposition 3.4. Both the repository and this manuscript are released under CC0 1.0 Universal.

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